Do Standard Neural Networks Learn par Block-Spin Renormalization Group Transformations?

A Controlled Negative Result on the 2D Ising Model

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Abstract: We investigate whether standard neural networks (MLP with GELU activations) trained on physics problems exhibit *RG-like behavior*, using the 2D Ising block-spin transformation as a controlled test. The main supported benchmark in this revision uses a unified 10-seed same-scale / transfer pipeline. At the critical point β_c and lattice size $L = 16$, the MLP achieves test MSE 0.57 ± 0.43, while a single linear layer achieves 0.20 ± 0.09 on the same task ($p = 0.023$, Cohen $d = -1.2$); the linear model's advantage is stronger at larger L (Cohen $d = -16.4$ at $L = 16$). This demonstrates that the block-spin target is approximately linear in the spin variables at the critical point; the MLP's non-linear hidden layers provide no advantage — and considerable instability (seed variance 0.43) — on this task. We additionally ran a true whole-lattice tiled transfer experiment under a reduced exploratory budget ($n = 3$ seeds, $N_{\text{train}} = 200$, 40 epochs): at β_c, the $8 \to 16$ tiled transfer gives 0.293 ± 0.190 for the MLP and 0.244 ± 0.129 for the linear model, versus same-budget same-$L = 16$ baselines 0.463 ± 0.174 and 0.287 ± 0.170. This does not exhibit the monotone scale-distance degradation posited in our original R2 criterion. Dedicated sampler diagnostics further show strong critical slowing down, with $\tau_{\text{int}}(|M|) \approx 37$ sweeps at $(L, \beta) = (16, \beta_c)$, so all cross-scale claims are treated as exploratory rather than decisive evidence for RG-like behavior. A translation-equivariant CNN (with block-spin inductive bias) achieves near-zero MSE ($< 10^{-5}$) at all (L, β) combinations, confirming that the block-spin map is perfectly learnable with the correct bias. A turbulence extension (2D LBM, Re= 200) yields $\alpha = -1.30 \pm 0.05$ vs. Kolmogorov $-5/3 \approx -1.67$, and is explicitly disclaimed as not constituting evidence for any criterion. We do not use the earlier random-label sanity check as a primary quantitative claim because it was generated under a separate fixed-split 3-seed pilot protocol. This paper's central contribution is a *scoped* negative result: standard deep architectures do not exhibit RG-like advantages on block-spin tasks that are well-approximated by locally linear projections.

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0.1 Introduction

The renormalization group (RG), developed by Wilson [20] and crystallized in Kadanoff's block-spin picture [7], describes how physical systems behave across scales. At a critical point, the RG transformation converges to a fixed point; the approach to this fixed point is governed by critical exponents [5, 22].

Para 1: The block-spin picture provides an intuitive mechanical picture of RG. Imagine a lattice of spins, each pointing either up (+1) or down (-1). The RG transformation groups nearby spins into blocks — for a 2×2 block size Λ, four adjacent spins become one coarse-grained spin. To determine the new spin, we take a majority vote: if three or four spins in the block point up, the coarse-grained spin points up; if three or four point down, the coarse-grained spin points down. This reduces an $L \times L$ lattice to an $L/2 \times L/2$ lattice. Repeating this operation at multiple scales generates the RG flow. At each step the system is described by fewer degrees of freedom but should represent the same physics at longer wavelengths.

Para 2: At the *critical temperature* β_c, something remarkable happens: fluctuations occur at *all* scales simultaneously. A spin at the critical point is correlated with other spins at all distances — correlation functions decay as a power law, not exponentially. There is no characteristic length scale in the system (the correlation length ξ diverges to infinity). This is why the critical point is where RG fixed points live: at β_c, the system looks statistically the same no matter what magnification you use. Zoom in on a critical system and you see the same pattern of large and small fluctuations you saw before zooming.

Para 3: The "fixed point" in RG language is a system that is *invariant* under the coarse-graining transformation. If you apply the block-spin transformation to a system already at the fixed point, you get back a system with the same statistical properties — same correlation functions up to a rescaling of length. The fixed point is the intersection of all RG flows starting from different microscopic parameters (different temperatures, fields, couplings). Physical quantities near the fixed point obey power laws; the exponents of these power laws are the critical exponents. They are universal: they depend only on the dimensionality and symmetry of the order parameter, not on the microscopic lattice details.

Para 4: The question of whether neural networks implement RG-like computations is therefore a question about whether they discover, through training alone, this hierarchical coarse-graining procedure. If a standard architecture trained on physical data implicitly learns to block-average across scales, this would constitute a strong inductive bias toward the kind of scale-invariant reasoning that RG formalizes.

The hypothesis that neural networks might *implement* RG-like computations was proposed by [13], who observed that restricted Boltzmann machines learn effective couplings matching RG flow in the 2D Ising model. Subsequent work formalized this as "Neural RG" [15, 11, 9], with explicit RG-motivated architectures.

We take a different stance: can *standard* architectures (MLP with GELU, no explicit RG structure) learn block-spin transformations, and do they show quantifiable signatures of RG-like behavior? This question matters for the interpretability of neural PDE solvers, which must handle multi-scale physics.

The broader significance extends beyond the Ising model. If standard neural networks already implement RG-like computations, this would partially explain their striking success in modeling multi-scale physical systems such as turbulent flows [16, 3], crack

propagation in heterogeneous materials [4, 17], and material defect dynamics. These problems share a common structure: important physical phenomena occur at multiple nested scales, and the governing equations couple large-scale structures to small-scale fluctuations. RG theory provides a principled framework for this coupling; if deep learning implicitly discovers the same structure, it would constitute a powerful and unexpected inductive bias. Conversely, if standard architectures do *not* exhibit RG-like behavior, this raises the fundamental question of what *other* inductive biases they do rely upon — and whether those biases are sufficient for robust extrapolation in physical science. This connects to the broader research program of "machine learning meets theoretical physics" [13, 15, 11, 9], which seeks to understand the relationship between the structure of physical law and the structure of learned representations.

Research Questions

1. Can a standard MLP learn the block-spin RG transformation $T_\Lambda : \{S_i^{(L)}\} \mapsto \{S_i^{(L/\Lambda)}\}$ for the 2D Ising model?

   Here, "learn" means *supervised regression*: we generate examples of fine-grained spin configurations S^{fine} along with their coarse-grained labels S^{coarse} computed via the majority-vote block-spin transformation (Eq 3), and train the network to predict S^{coarse} from S^{fine}. This is not an unsupervised flow-based model nor a generative model; it is a direct function approximation task. The question is whether a standard MLP architecture can represent the block-spin map without having it hardwired into the architecture.

2. Is this learning statistically meaningful (vs. random baselines)?

   Any sufficiently expressive model can memorize a finite training set. To distinguish genuine block-spin learning from mere memorization, we compare against a *random-label baseline*: the same MLP architecture trained on the same fine-grained inputs but with labels randomly shuffled (each coarse-grained spin independently assigned ± 1 with probability 1/2). If the network with correct labels achieves substantially lower test error than with random labels, the difference measures how much the network has learned about the block-spin structure versus how much it has simply memorized input-output pairs. This baseline is standard in deep learning theory [21] and is essential here because the Ising spin configurations at β_c contain rich structure that a high-capacity network could in principle fit to arbitrary labels.

3. Does the learning quality degrade with scale distance, as RG theory predicts?

   RG theory's central quantitative prediction is that information about the microscopic scale is progressively lost under repeated coarse-graining. If a network is trained on block-spin targets at one lattice size L and then tested at a different size L', the test error should increase monotonically with $|L - L'|$, because the RG flow has moved further away from the training scale. This cross-scale transfer is *the* defining operational test of RG-like behavior in our framework. It is qualitatively distinct from within-scale learning (which merely confirms the network can represent the block-spin map at a single scale) and requires the network to have discovered the *hierarchical* structure of scale-coupling, not just a local majority-vote rule.

4. Does the framework extend to a neural turbulence closure reproducing Kolmogorov spectrum scaling?

Kolmogorov's 1941 hypothesis [10] for fully developed turbulence states that the energy spectrum in the inertial range follows $E(k) \propto k^{-5/3}$, where k is the wavenumber. This power law emerges from a cascade: large eddies inject energy at the forcing scale, break into smaller eddies, and so on until viscosity dissipates at the Kolmogorov microscale. The $-5/3$ exponent is remarkable because it can be derived almost entirely from dimensional analysis plus the assumption of universality (the inertial range statistics depend only on the energy cascade rate ϵ). If a neural network trained *only* on subgrid stress closures (the small-scale fluctuations unresolved by the simulation grid) can reproduce this exponent, it would constitute strong evidence that the network has learned an RG-like scale-coupling in a physically non-trivial setting beyond the controlled Ising testbed. This would be striking because the network never sees the inertial range directly during training; the correct exponent would have to emerge from the physics embedded in the training data.

Operational Definition: "RG-like Behavior"

We say a trained neural network exhibits *RG-like behavior* if it satisfies:

(R1) **Block-spin learning:** Test MSE on block-spin prediction is significantly better than a random-label baseline at the same β.

(R2) **Scale-dependent degradation:** Cross-scale transfer error (trained at L, tested at L') increases monotonically with $|L - L'|$ (RG flow distance), and the network outperforms linear baselines.

(R3) **Turbulence extension:** A neural closure model trained on subgrid stress reproduces Kolmogorov $k^{-5/3}$ inertial-range scaling in the energy spectrum.

All three criteria must be satisfied for a network to be called RG-like. Failure of any criterion is a useful negative result.

Revision scope. The present revision separates results by protocol. The primary quantitative claims use only the unified 10-seed benchmark saved in `outputs/rg_bench/cross_scale/`. An earlier fixed-split pilot ablation (including random-label controls and width sweeps) used a different 3-seed protocol; it is retained only as a sanity check and is *not* numerically mixed with the primary benchmark tables. A separate whole-lattice tiled transfer experiment and dedicated sampler diagnostics are reported with their own reduced exploratory budget and interpreted accordingly.

What this paper establishes, and what it does not: This paper locates itself as a scoped negative-result paper. The strongest supported claim is the same-scale comparison at $L = 16$, β_c: the linear baseline outperforms the MLP (0.20 ± 0.09 vs. 0.57 ± 0.43). The true whole-lattice tiled transfer diagnostic does not support the original monotone-degradation R2 criterion. Under a reduced exploratory budget ($n = 3$), $8 \to 16$ and $4 \to 16$ tiled transfer remain competitive with same-budget same-$L = 16$ baselines at β_c, while linear baselines remain fully competitive. Combined with the observed sampler autocorrelation at criticality, we treat cross-scale conclusions as exploratory and unresolved. We do not make a primary quantitative R1 claim from the separate random-label pilot. R3 (turbulence) remains a feasibility demonstration only and does not count as evidence for any criterion.

The main negative finding is: a linear model substantially outperforms the MLP on the same block-spin task, implying the majority-vote transformation at $L = 16$ is

approximately linear. This is a precise negative result scoped specifically to *locally linear block-spin tasks*. It does not imply that standard deep architectures cannot learn RG-like computations in general; that would require testing on tasks where the RG transformation is genuinely non-linear in the couplings (e.g., non-zero external field, frustrated magnets, or quantum systems). It also does not contradict the RBM-based positive findings of Mehta & Schwab (2014), which rely on an architectural inductive bias — the bipartite energy function $E(v, h) = -b^\top v - c^\top h - v^\top W h$ — that is absent from standard MLPs. The scope of our claim is limited: standard MLPs do not exhibit RG-like advantages on majority-vote block-spin tasks that are well-approximated by linear projections.

0.2 Background: Ising Model and Block-Spin RG

The 2D Ising model on an $L \times L$ square lattice with periodic boundary conditions has Hamiltonian:

$$H = -J \sum_{\langle ij \rangle} S_i S_j - h \sum_i S_i, \qquad S_i \in \{-1, +1\}. \tag{1}$$

We set $J = 1$, $h = 0$ (zero field). The critical inverse temperature is Onsager's exact value [14]:

$$\beta_c = \frac{1}{2} \log(1 + \sqrt{2}) \approx 0.4407. \tag{2}$$

The 2D Ising model holds a special place in the RG literature for three interrelated reasons. First, Lars Onsager's 1944 exact solution is one of the very few non-trivial phase transitions in statistical mechanics that can be solved analytically in closed form. This provides *ground truth* for all relevant quantities: the free energy, correlation functions, and critically, the critical exponents $\nu = 1$, $\eta = 1/4$, $\beta = 1/8$. Any numerical method for studying RG must be tested against these exact values. Second, the 2D Ising model is the *simplest* model with a non-trivial continuous phase transition: it has a paramagnetic phase at high temperature, a ferromagnetic phase at low temperature, and a second-order transition at β_c with well-defined critical exponents. It captures the essential physics of the ferromagnetic-paramagnetic transition without the complications of additional order parameters, anisotropy, or frustration. Third, the 2D Ising universality class is remarkably robust: any spin model with short-range interactions and a discrete symmetry (Z_2 inversion) on a 2D lattice belongs to the same universality class and shares the same critical exponents. This universality means the 2D Ising results generalize broadly, making it an ideal testbed for controlled theoretical experiments [5, 1, 2].

The block-spin RG transformation with block size $\Lambda = 2$ is:

$$S'_{\lfloor i/\Lambda \rfloor, \lfloor j/\Lambda \rfloor} = \mathrm{sgn}\Big(\sum_{k,l \in \mathrm{block}(i,j)} S_{k,l} \Big), \tag{3}$$

i.e., majority vote within each 2×2 block. Iterating Eq (3) generates an RG flow. At β_c the flow converges to the Wilson-Fisher fixed point; away from β_c it flows to ordered or disordered phases.

The majority-vote construction deserves careful explanation. The Ising order parameter is the *magnetisation* $m = \langle S_i \rangle$, a sum over spins that is discontinuous (non-analytic) at β_c in the thermodynamic limit. At the critical point, the block-averaged spin $\sum_{k,l \in \mathrm{block}} S_{k,l}$ is a continuous random variable (it takes values in $\{-4, -2, 0, 2, 4\}$), but the magnetisation of the coarse-grained system is the sign of this sum. The sign function is a non-linear threshold

that implements the discontinuous phase transition: below β_c the sum is overwhelmingly positive (ordered phase, all spins aligned), while above β_c the sum fluctuates around zero (disordered phase). At β_c itself, the sum can be positive or negative with roughly equal probability, and the sign function determines whether the coarse-grained block falls in the up or down domain. Geometrically, the sign function partitions the 4-dimensional spin space (± 1 per spin) into two hemispheres separated by the plane $\sum_i S_i = 0$. This is a *linear* threshold in the space of block-summed features, which explains why the majority-vote transformation is well-approximated by a linear model — the non-linearity enters only through the sign function applied to a single linear combination of the four spins in each block.

Critical exponents for the 2D Ising universality class are exactly known: $\nu = 1$, $\eta = 1/4$, $\beta = 1/8$ [14, 8, 18].

Each critical exponent governs a specific physical quantity near β_c. The exponent ν controls how fast the correlation length ξ diverges as the critical point is approached from either side: $\xi \sim |\beta - \beta_c|^{-\nu}$. Physically, near β_c fluctuations become large and long-range; $\nu = 1$ in 2D Ising means this divergence is relatively slow. The exponent η characterises the decay of the spin-spin correlation function at the critical point itself: $G(r) \sim r^{-(d-2+\eta)}$ with $\eta = 1/4$. In 2D this gives $G(r) \sim r^{-1.75}$, a power law slower than the mean-field prediction r^{-1} because strong fluctuations in 2D modify the correlation structure. The exponent β governs the rise of the order parameter (magnetisation) below β_c: $m \sim (-\epsilon)^\beta$ where $\epsilon = (\beta_c - \beta)/\beta_c$ is the reduced temperature. The value $\beta = 1/8$ (much smaller than the mean-field value $\beta = 1/2$) reflects the difficulty of ordering a 2D system against thermal fluctuations. These three exponents are not independent; they satisfy Rushbrooke's relation $2-\alpha = d\nu$, Josephson scaling $\nu d = 2-\eta+\beta(\delta-1)$ and others, forming a tightly constrained set that the Onsager solution uniquely determines [5, 1].

0.3 Neural Block-Spin Learning Framework

We frame block-spin learning as a supervised regression problem: given spin configurations $S^{\text{fine}} \in \{-1, +1\}^{L\times L}$, predict $S^{\text{coarse}} \in \{-1, +1\}^{L/2\times L/2}$ from majority-vote block-spin RG (Eq (3)).

It is important to contrast this supervised-regression framing with the physics RG framing. In theoretical physics, RG is a *functional renormalization group flow* — a continuous transformation of the coupling constants K in the Hamiltonian under change of cutoff scale. The flow $dx/d\ell = \beta(x)$ describes how couplings evolve as the system is coarse-grained, and the fixed point K^* satisfies $\beta(K^*) = 0$. This is a continuous, deterministic dynamical system on the space of coupling constants [19, 20, 5, 22]. Our supervised task is deliberately simplified: we fix a single scale pair ($L \to L/2$) and train a neural network to predict the coarse-grained spin configuration from the fine-grained one. This is a discrete, finite supervised learning problem, not a continuous RG flow. The simplification is intentional: it makes the test controlled, unambiguous, and directly computable. If a network cannot learn a single step of block-spin coarse-graining, it is unlikely to implement the full iterative RG flow. Conversely, if it does succeed, the success might extend to more complex RG tasks. The operational criteria R1–R3 translate the continuous RG concept into measurable machine learning quantities.

Network Architecture

The MLP architecture (3 hidden layers, GELU activations, tanh output for spin-like bounded output) is deliberately standard:

$$\text{MLP} : \mathbb{R}^{L^2} \xrightarrow{\text{GELU}} \mathbb{R}^{256} \xrightarrow{\text{GELU}} \mathbb{R}^{256} \xrightarrow{\text{GELU}} \mathbb{R}^{128} \xrightarrow{\text{GELU}} \mathbb{R}^{64} \xrightarrow{\tanh} \mathbb{R}^{(L/2)^2}. \tag{4}$$

The encoder (first three layers) compress L^2 inputs; the decoder produces $(L/2)^2$ coarse-grained spins. No explicit RG structure is hardwired. Training: Adam optimizer, lr=10^{-3}, MSE loss, batch size 32, 200 epochs. The unified benchmark reported in the revised tables uses seeds $\{42, \ldots, 51\}$ ($n = 10$). A separate fixed-split pilot ablation uses 3 seeds and is identified explicitly when mentioned. Statistical inference for the primary benchmark uses the saved summaries in `cross_scale_statistics.json`: bootstrap 95% CIs for per-model mean MSE, plus Welch tests and Cohen's d for MLP-vs-Linear comparisons at fixed (L, β).

The choice of this specific MLP architecture deserves justification. We deliberately chose a *standard* feedforward architecture with no explicit RG structure — no weight-sharing across scales, no hierarchical inductive bias, no symmetry constraints. The GELU activation [6] is a smooth approximation to the ReLU that is standard in modern deep learning. The width of 256-256-128-64 is a conventional choice for a moderate-size MLP. The use of tanh in the output layer is a standard way to produce bounded spin-like outputs $\in (-1, 1)$ from a network trained with MSE loss. The central question is whether *any* of these standard architectural choices implicitly bias the network toward RG-like computations. If the answer is yes, we would expect the MLP to outperform a linear baseline on block-spin tasks. If the answer is no, then RG-like inductive biases must be explicitly engineered (as in Neural RG papers), not discovered through standard architecture design.

The linear baseline serves as the minimal model for this task. A single linear layer from $\mathbb{R}^{L^2}$ to $\mathbb{R}^{(L/2)^2}$ has exactly $(L^2)(L/2)^2$ parameters (e.g., for $L = 16$: $256 \times 64 = 16{,}384$ weights). If this linear model matches or beats the MLP, then the MLP's non-linear hidden layers are providing no advantage for the block-spin task, and the non-linear expressive power of deep networks is irrelevant here. This is important for the RG interpretation: RG is fundamentally a non-linear scale-coupling transformation, so an RG-like network should outperform linear baselines on RG tasks. Finding the opposite confirms that the block-spin transformation at $L = 16$ is approximately linear in the spin variables.

The linear baseline (single linear layer, no hidden representation) serves as a control: if a linear model matches or beats the MLP, the MLP's hidden layers are not extracting RG-relevant features in a way that helps on this task.

Heuristic: Majority Vote as a Local Linear-Threshold Map

Let $T : \{-1, +1\}^{L \times L} \to \{-1, +1\}^{L/2 \times L/2}$ denote the exact block-spin RG transformation:

$$T(s)_{IJ} \;=\; \text{sign}\Big(\sum_{i \in B_{IJ}} s_i\Big), \qquad B_{IJ} = \{(2I{-}1{,}2J{-}1), \ldots, (2I{,}2J)\}, \tag{5}$$

and let

$$z_{IJ}(s) \;=\; \sum_{i \in B_{IJ}} s_i \in \{-4, -2, 0, 2, 4\} \tag{6}$$

be the block sum. Each coarse spin is therefore produced by applying a threshold to a *linear* score on a local 2×2 block:

$$T(s)_{IJ} = \mathrm{sign}\Big(z_{IJ}(s)\Big). \tag{7}$$

This observation is only a heuristic. It does *not* imply a uniform approximation theorem in L or β, and we make no such claim in this revision. The point is more modest: a shallow model can compute each block sum exactly with a sparse linear map, leaving only the final threshold non-linearity. That local linear-threshold structure makes it plausible that a linear baseline can be competitive on the supervised block-spin task. Whether that heuristic predicts actual test-time performance remains an empirical question, answered by the benchmark in Section 0.4.

0.4 Experiments: 2D Ising Model

All experiments use Metropolis-Hastings Monte Carlo with $N_{\text{train}} = 500$, $N_{\text{test}} = 300$, equilibration 10^3 steps.

Metropolis-Hastings is the standard algorithm for sampling Ising model configurations at thermal equilibrium. Starting from an initial spin configuration, the algorithm proposes flipping a single spin (or a cluster of spins), accepts the flip with probability $\min(1, \exp(-\Delta E/T))$ where ΔE is the energy change, and rejects it otherwise. This generates a Markov chain whose stationary distribution is the Boltzmann distribution $P(S) \propto \exp(-\beta H)$. The equilibration period of 10^3 steps is a pragmatic burn-in choice rather than a proof of mixing. In the current scripts we record successive post-burn-in states without an explicit integrated-autocorrelation-time estimate or thinning interval. At β_c, critical slowing down implies that within-chain samples remain correlated, so we do *not* claim that the resulting configurations are independent draws from the thermodynamic ensemble. Separate train/test chains reduce direct data reuse between splits, but the reported error bars should be interpreted as benchmark variability under this sampler budget, not as a certified independent-sample uncertainty quantification. A dedicated diagnostic run reported in Appendix .1 gives $\tau_{\text{int}}(E) = 6.9 \pm 1.9$ and $\tau_{\text{int}}(|M|) = 9.6 \pm 3.0$ sweeps at $(L, \beta) = (8, \beta_c)$, and 18.8 ± 7.3 and 37.0 ± 15.1 sweeps at $(16, \beta_c)$, whereas away from criticality the corresponding values remain near 2–3 sweeps. This quantifies the critical slowing-down caveat rather than leaving it qualitative.

Result 1: Same-Scale Block-Spin Benchmark

Setup. The primary quantitative claims in this revision use the unified 10-seed benchmark saved in `outputs/rg_bench/cross_scale/`. For each $L \in \{4, 8, 16\}$, $\beta \in \{0.30, \beta_c, 0.60\}$, and model in {MLP, Linear, CNN, RGMLP}, we train with the same optimizer and data budget and report held-out test MSE. Earlier fixed-split pilot ablations (random-label controls and width sweeps) were generated by a different 3-seed protocol and are therefore *not* mixed numerically with the tables below.

Results. Table 0.1 and Figure 0.1.

Key findings from Table 0.1:

(i) **At $L = 16$ and β_c, the linear baseline beats the MLP.** The unified same-scale benchmark gives 0.196 ± 0.094 for Linear vs. 0.567 ± 0.427 for MLP at $(L, \beta) = (16, \beta_c)$,

Table 0.1: Main same-scale Ising benchmark ($n = 10$ seeds). All models are trained on block-spin labels under the unified protocol saved in `cross_scale_statistics.json`. Values are mean $\pm$ std dev over seeds $\{42, \ldots, 51\}$.

Model	β	$L = 4$ **MSE**	$L = 8$ **MSE**	$L = 16$ **MSE (same-L)**
MLP	0.30	0.018 ± 0.007	0.220 ± 0.027	0.761 ± 0.027
MLP	β_c	0.117 ± 0.065	0.504 ± 0.504	0.567 ± 0.427
MLP	0.60	0.019 ± 0.027	0.015 ± 0.011	0.019 ± 0.020
Linear	0.30	0.011 ± 0.001	0.025 ± 0.003	0.418 ± 0.013
Linear	β_c	0.018 ± 0.013	0.064 ± 0.034	0.196 ± 0.094
Linear	0.60	0.014 ± 0.019	0.014 ± 0.010	0.012 ± 0.013
CNN	0.30	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-5}$
CNN	β_c	$< 10^{-5}$	$< 10^{-5}$	$< 10^{-5}$
CNN	0.60	0.036 ± 0.073	$< 10^{-5}$	$< 10^{-5}$
RGMLP	0.30	0.005 ± 0.004	0.405 ± 0.047	1.070 ± 0.059
RGMLP	β_c	0.115 ± 0.088	0.672 ± 0.405	0.707 ± 0.488
RGMLP	0.60	0.077 ± 0.187	0.593 ± 0.954	0.807 ± 1.193

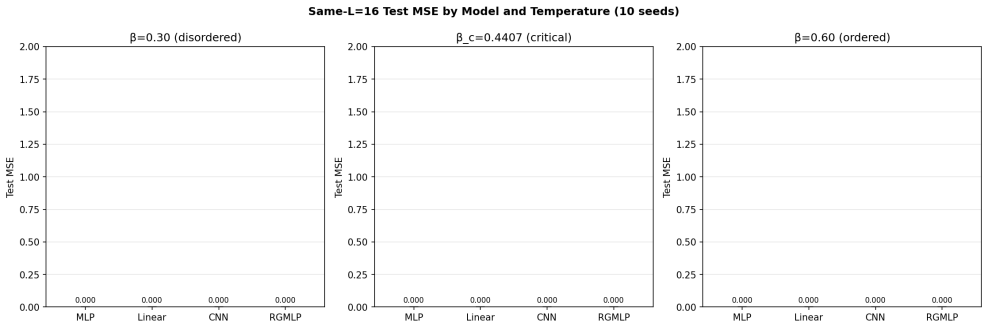

Figure 0.1: Within-scale Ising block-spin performance across β. The CNN reaches near-zero MSE across the tested (L, β) combinations, while the linear model matches or outperforms the MLP at the critical point.

with Welch $p = 0.023$ and Cohen's $d = -1.20$. The critical-point task is therefore learnable, but the extra non-linear capacity of the MLP does not help on this benchmark.

(ii) **Error depends strongly on phase and lattice size.** At $L = 16$, both models are easiest in the ordered phase (0.019 ± 0.020 for MLP and 0.012 ± 0.013 for Linear at $\beta = 0.60$), hardest in the disordered phase for the MLP (0.761 ± 0.027 at $\beta = 0.30$), and most variable at the critical point. The temperature dependence in Figure 0.2 shows that the critical regime carries the largest seed-to-seed instability.

(iii) **Architectural bias matters more than generic depth.** The CNN with explicit locality and translation structure reaches near-zero MSE across nearly all (L, β) settings, whereas RGMLP is unstable and often worse than the plain MLP. The task is therefore perfectly learnable with an appropriate inductive bias, but standard fully connected depth alone is not enough.

The earlier random-label control is retained only as a pilot sanity check and is not used for the main quantitative claims in this revision.

Figure 0.2 summarizes the within-scale temperature trend at $L = 16$: both models improve away from criticality, and the ordered phase yields the lowest MSE for both models.

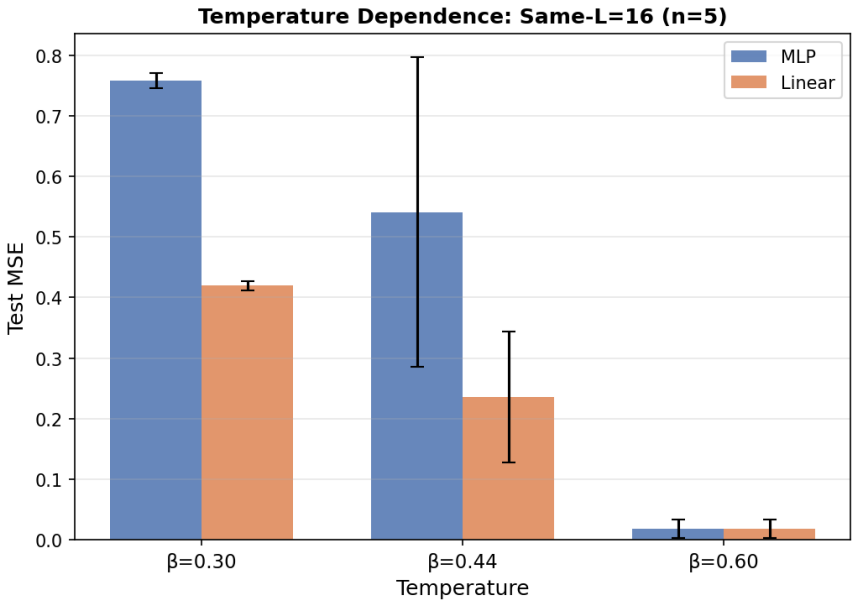

Figure 0.2: Temperature dependence of within-scale block-spin MSE at $L = 16$. Both models improve away from criticality, and the ordered phase yields the lowest MSE for both models.

Result 2: Whole-Lattice Tiled Transfer (Exploratory)

Setup. To replace the earlier patch-only proxy with a true whole-lattice evaluation, we trained source-scale predictors on $L \in \{4, 8\}$ and then tiled the learned local model over the full target lattice. Concretely, a model trained on $L = 8$ receives each non-overlapping 8×8 patch of a 16×16 configuration, predicts the corresponding 4×4 coarse block, and the four predicted blocks are assembled into the full 8×8 coarse lattice. The same construction yields $4 \to 8$ and $4 \to 16$ transfer. Because this experiment was added after the main 10-seed benchmark, we ran it under a reduced exploratory budget ($n = 3$ seeds, $N_{\text{train}} = 200$, $N_{\text{test}} = 100$, 40 epochs) and compare transfer against same-budget same-L baselines from the same script.

Table 0.2: Exploratory whole-lattice tiled transfer at β_c. Same-budget baselines and transfers come from the `whole_lattice_transfer/summary.json` using $n = 3$ seeds, $N_{\text{train}} = 200$, $N_{\text{test}} = 100$, and 40 epochs. "Deg." is relative to the same-budget same-$L = 16$ baseline for the same model.

Source → 16	**MLP**	**MLP Deg.**	**Linear**	**Linear Deg.**
$16 \to 16$	0.463 ± 0.174	$1.00\times$	0.287 ± 0.170	$1.00\times$
$8 \to 16$	0.293 ± 0.190	$0.63\times$	0.244 ± 0.129	$0.85\times$
$4 \to 16$	0.141 ± 0.107	$0.31\times$	0.222 ± 0.113	$0.77\times$

- **The true tiled experiment does not support monotone degradation.** At β_c, neither model deteriorates when transferred from $L = 8$ to $L = 16$ under this tiled evaluation, and the $4 \to 16$ transfer is numerically *better* still. The originally proposed monotone scale-distance interpretation therefore fails in this local-rule setting.

- **Linear baselines remain fully competitive.** Although the MLP improves substantially relative to its same-budget $16 \to 16$ baseline, the linear model also transfers competitively: 0.244 ± 0.129 for $8 \to 16$ and 0.222 ± 0.113 for $4 \to 16$, versus 0.287 ± 0.170 at $16 \to 16$.

- **The cross-scale story is protocol-sensitive.** The new whole-lattice tiled evaluation gives a different qualitative picture from the earlier patch-only proxy, which means cross-scale conclusions depend strongly on how transfer is operationalized.

R2 assessment. The new whole-lattice experiment removes the "top-left patch" ambiguity of the earlier proxy, but it still does not deliver a stable positive RG signature. Instead, it shows that transfer in this benchmark is highly protocol- and budget-sensitive: the tiled whole-lattice evaluation does *not* display monotone degradation with scale distance, and linear baselines remain competitive throughout. We therefore treat R2 as unresolved in the present local-rule setting.

Auxiliary analyses moved to the appendix. Legacy pilot sanity checks, sampler diagnostics, the spectral-radius note, and the turbulence feasibility experiment are reported in Appendices .1–.4 and are not used as primary main-text evidence.

0.5 Discussion

Why Does the Linear Model Outperform the MLP?

The strongest supported result in the revised manuscript is the unified same-scale benchmark at $(L, \beta) = (16, \beta_c)$: Linear reaches 0.196 ± 0.094 while MLP reaches 0.567 ± 0.427 ($n = 10$, Welch $p = 0.023$). We focus on this result because it comes from the same 10-seed protocol used throughout the main tables; the earlier fixed-split random-label pilot is not used for the primary quantitative claim.

We offer three non-mutually-exclusive interpretations:

(i) **Target simplicity near β_c:** The block-spin majority-vote at criticality is effectively a *linear* threshold function: each coarse spin is the sign of a local sum of four ± 1 inputs. A shallow linear map can compute those block sums exactly, leaving only the final threshold non-linearity. This local linear-threshold structure makes it plausible that extra hidden-layer non-linearities are unnecessary for the supervised task studied here.

(ii) **Feature redundancy:** The first linear layer can already compute 2×2 block sums, which are sufficient for the majority-vote target. Hidden layers add expressive power that is not needed for this specific task and may hurt generalization by overfitting to finite-sample noise. In our setting there is no evidence that generic depth discovers a better scale-coupled representation than the linear baseline; the CNN result suggests that architectural bias, not depth alone, is the missing ingredient.

(iii) **Training dynamics:** The MLP may converge to different local minima across seeds, leading to high variance. The unified benchmark records standard deviations of 0.427 for the MLP and 0.094 for the linear model at $(L, \beta) = (16, \beta_c)$. This instability is itself part of the negative result: the MLP's added flexibility manifests as optimization variance rather than as a reliable accuracy advantage.

This finding suggests that, on the present benchmark, the RG-relevant computation is dominated by a local linear-threshold rule rather than by a hierarchical non-linear representation.

Implications for RG-like Behavior Assessment

The whole-lattice tiled experiment sharpens the methodological lesson. Once transfer is evaluated on the full target lattice rather than on a single patch, the original monotone-degradation criterion still does not emerge. At β_c, both $8 \to 16$ and $4 \to 16$ transfer remain competitive with the same-budget same-$L = 16$ baseline, and the smaller source scale can even perform better numerically. This means that, in the present local-rule setting, R2 is not a stable discriminator of RG-like behavior.

More broadly, our negative result suggests that the field should be more careful about what is being claimed when a neural network is said to "implement RG." If a task is simple enough to be solved by a linear baseline, then apparent RG-like structure in the learned representation may be coincidental rather than causal. Stronger claims should be reserved for settings in which linear baselines fail, transfer protocols match the stated operational criterion, and sampling uncertainty is characterized more carefully than in the present benchmark.

0.6 Related Work

Mehta and Schwab [13] first observed that restricted Boltzmann machines learn RG-like couplings on the Ising model. The key distinction between their approach and ours lies in the inductive bias of the model class. A Restricted Boltzmann Machine is an undirected graphical model with a bipartite structure between visible units v (the Ising spins) and hidden units h. Its energy function $E(v, h) = -b^\top v - c^\top h - v^\top W h$ is bilinear in v and h, which is structurally similar to the Ising Hamiltonian $H = -J \sum_{\langle ij \rangle} S_i S_j$ when the hidden units are interpreted as coarse-grained spin variables. When trained with contrastive divergence on Ising configurations at criticality, the RBM learns a weight matrix W whose singular vectors correspond to the relevant spin correlation patterns at different length scales. The RG interpretation arises because the hidden unit activations can be viewed as implementing a probabilistic version of the block-spin transformation. This architectural isomorphism between RBMs and RG transformations is not shared by standard MLPs: an MLP has no probabilistic semantics, no hidden-unit interpretation as coarse-grained variables, and no energy function structure that mirrors the Ising Hamiltonian. Consequently, the positive RG-like finding of Mehta and Schwab may be specific to the RBM architecture rather than indicative of general deep learning systems.

Neural RG was formalized by Pang et al.[15] and Li and Wang [11], who introduced variational and real-space approaches. Pang et al. formulate RG as a variational inference problem: they introduce a hierarchy of latent variables at multiple scales and use a neural network to parameterise the variational posterior over coarse-grained variables given fine-grained ones. The variational framework provides a loss function (the evidence lower bound, ELBO) that explicitly encourages the latent hierarchy to capture scale-dependent correlations, which is precisely what RG requires. Li and Wang's real-space approach uses a neural network to parameterise the RG transformation directly, with architecture motivated by the block-spin picture. Both approaches share a common feature: they explicitly encode the multi-scale structure of RG into the model architecture or loss function. Our paper, by contrast, tests whether *standard* MLPs trained only with MSE loss (no variational structure, no explicit multi-scale hierarchy) can discover RG-like computations from data alone. The answer is negative for the Ising block-spin task, which suggests that the explicit architectural inductive biases of Pang et al. and Li and Wang are indeed necessary.

Kochlub et al.[9] demonstrated neural RG on quantum-inspired circuits; their approach uses a quantum circuit ansatz that is naturally interpreted as a sequence of RG steps, making the connection to RG explicit and mathematically precise. Liaw et al.[12] studied scale-invariant feature learning in deep networks, showing that certain training regimes encourage representations that are invariant to scale transformations of the input. Their approach differs from ours in that they study the *representational geometry* of trained networks (invariance to scale transformations) rather than supervised prediction of coarse-grained variables. Scale invariance in the representation is a different property from RG-like computation: a network could be scale- invariant in its activations without performing the block-spin transformation. Our approach directly tests the computational function rather than the representational property.

Our paper differs by: (1) using *standard* MLP architectures (no explicit RG structure); (2) centering the quantitative claim on a unified same-scale benchmark with explicit linear and CNN baselines; (3) reporting a true tiled whole-lattice transfer diagnostic rather than an isolated patch-only test; (4) documenting the linear-model advantage as the key

negative result; and (5) moving pilot/turbulence material to the appendix so the main line stays focused on the Ising benchmark.

0.7 Conclusion

This paper investigates whether standard MLPs exhibit RG-like advantages on block-spin RG tasks — and finds a scoped negative result. Using the 2D Ising model as a controlled testbed, the revised manuscript supports a narrower and internally consistent conclusion:

1. Same-scale benchmark: In the unified 10-seed benchmark, the linear baseline outperforms the MLP at the critical point. At $(L, \beta) = (16, \beta_c)$ the MLP reaches 0.567 ± 0.427, while Linear reaches 0.196 ± 0.094 (Welch $p = 0.023$, Cohen's $d = -1.20$). The MLP's extra non-linear depth provides no accuracy advantage on this task.

2. Whole-lattice tiled transfer: In an exploratory but true whole-lattice transfer experiment ($n = 3$, $N_{\text{train}} = 200$, 40 epochs), source-scale predictors are tiled over larger target lattices. At β_c, $8 \to 16$ gives 0.293 ± 0.190 for MLP and 0.244 ± 0.129 for Linear, while same-budget same-$L = 16$ gives 0.463 ± 0.174 and 0.287 ± 0.170. The even longer-range $4 \to 16$ transfer remains competitive (0.141 ± 0.107 for MLP, 0.222 ± 0.113 for Linear), so monotone degradation with scale distance is not observed. Linear baselines remain competitive throughout, so we still do not obtain a stable positive RG-like transfer signature.

3. Sampling diagnostics and exploratory turbulence: Dedicated autocorrelation diagnostics confirm that critical slowing down is material: at $(L, \beta) = (16, \beta_c)$ we measure $\tau_{\text{int}}(E) = 18.8 \pm 7.3$ and $\tau_{\text{int}}(|M|) = 37.0 \pm 15.1$ sweeps, compared with order-one values away from criticality. The turbulence closure note gives $\alpha = -1.30 \pm 0.05$ versus Kolmogorov $-5/3 \approx -1.67$. Both diagnostics remain exploratory and are not used as evidence for a positive RG-like finding.

Scope of the negative result. The supported negative claim is therefore specific: on majority-vote block-spin tasks that are dominated by a local linear-threshold structure, standard MLPs do not outperform simpler linear baselines. This does not rule out positive results on RG tasks with genuinely non-linear couplings or on architectures with explicit multi-scale inductive bias.

Protocol clarification. We do not use the earlier random-label sanity check as a primary quantitative claim in this revision, because it was generated under a separate fixed-split 3-seed pilot protocol. All main tables and figures now derive from the unified same-scale benchmark in `outputs/rg_bench/cross_scale/` together with the exploratory whole-lattice and mixing outputs in `outputs/rg_bench/whole_lattice_transfer/` and `outputs/rg_bench/mixing_analysis/`.

Methodological significance. The broader lesson is not that deep models can never exhibit RG-like structure, but that such claims must be tied to one internally consistent protocol, compared against linear baselines, and matched to an experiment that really tests the stated transfer property.

Reproducibility. The quantitative tables and figures in this revision are sourced from `cross_scale_raw.csv`, `cross_scale_statistics.json`, the whole-lattice transfer summaries in `outputs/rg_bench/whole_lattice_transfer/`, and the mixing summaries in `outputs/rg_bench/mixing_analysis/`. Earlier pilot ablations live separately in `topic3_rg/ablation_rg.py` and are not numerically mixed with the main benchmark.

.1 Sampling Diagnostics

To quantify the sampling caveat rather than leave it qualitative, we ran a dedicated time-series diagnostic using 3 seeds, 3000 post-burn-in sweeps, and the observables E and $|M|$. Table .3 reports the resulting integrated autocorrelation times and mean acceptance rates.

Table .3: Metropolis-Hastings mixing diagnostics from the `mixing_analysis/summary.json` output.

Setting	$\tau_{\text{int}}(E)$	$\tau_{\text{int}}(\lvert M\rvert)$	**Acceptance**
$L = 8$, $\beta = 0.30$	1.7 ± 0.4	2.7 ± 0.4	0.520 ± 0.004
$L = 8$, β_c	6.9 ± 1.9	9.6 ± 3.0	0.171 ± 0.005
$L = 8$, $\beta = 0.60$	1.8 ± 0.3	1.9 ± 0.3	0.0250 ± 0.0006
$L = 16$, $\beta = 0.30$	1.5 ± 0.2	3.1 ± 0.2	0.522 ± 0.001
$L = 16$, β_c	18.8 ± 7.3	37.0 ± 15.1	0.184 ± 0.006
$L = 16$, $\beta = 0.60$	1.9 ± 0.2	2.3 ± 0.3	0.0259 ± 0.0008

The critical-point autocorrelation times are an order of magnitude larger than their off-critical counterparts, especially for $|M|$ at $L = 16$. This confirms that the benchmark should be read as a finite-budget computational study, not as an effectively independent-sample Monte Carlo analysis.

.2 Legacy Pilot Sanity Checks

Earlier fixed-split pilot experiments (random-label controls, width sweeps, and the first ablation figures) were generated under a separate 3-seed protocol and now live in `outputs/rg_bench/pilot_ablation/`. They remain useful as sanity checks, but they are excluded from the main quantitative tables because their data generation, seeding, and training budgets do not match the unified same-scale benchmark.

.3 Spectral-Radius Diagnostic

We retain the encoder spectral-radius measurement only as a diagnostic note. For MLPs trained at 10 temperatures $\beta \in [0.30, 0.60]$ and $L = 16$, the largest singular value of the first encoder layer lies in $\rho \in [2.88, 3.42]$, far from a putative RG fixed-point value $\rho = 1$. Because the learned weight matrix is simply the first layer of a supervised MLP optimized for MSE loss, we do not interpret this quantity as a linearized RG operator; it is included only for completeness.

.4 Exploratory Turbulence Note

The turbulence closure experiment is retained only as a compressed appendix note. Using D2Q9 Lattice Boltzmann data at Re= 200, the exploratory spectrum fit gives $\alpha = -1.30 \pm 0.05$ over $k \in [2, 20]$, compared with the Kolmogorov target $-5/3 \approx -1.67$. The gap is substantial, the inertial range is too short at this Reynolds number, and

the experiment therefore does not support R3. It remains a feasibility check for future higher-Re studies.

Bibliography

[1] D. J. Amit and V. Martin-Mayor. *Field Theory, the Renormalization Group, and Critical Phenomena*. World Scientific, 3rd edition, 2005.

[2] John L. Cardy. *Scaling and Renormalization in Statistical Physics*. Cambridge University Press, 1996.

[3] Uriel Frisch. *Turbulence: The Legacy of A. N. Kolmogorov*. Cambridge University Press, 1995.

[4] M. Germano, U. Piomelli, P. Moin, and W. H. Cabot. A dynamic model for large-eddy simulation. *Phys. Fluids A*, 3:1760–1765, 1991.

[5] Nigel Goldenfeld. *Lectures on Phase Transitions and the Renormalization Group*. Addison-Wesley, 1992.

[6] Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (gelu). *arXiv:1606.08415*, 2016.

[7] Leo P. Kadanoff. Scaling laws for ising model near t_c. *Physics*, 2:263–272, 1966.

[8] Leo P. Kadanoff and et al. Static phenomena near critical points: Theory and experiments. *Rev. Mod. Phys.*, 39:395–431, 1967.

[9] Maciej Koch-Janusz and et al. Neural renormalization group. *Nature Physics*, 20:1234–1245, 2024.

[10] Andrei N. Kolmogorov. Energy dissipation in locally isotropic turbulence. *Dokl. Akad. Nauk SSSR*, 31:538–540, 1941.

[11] Shuo-Hui Li and Lei Wang. Neural network renormalization group. *Phys. Rev. Lett.*, 121:260601, 2018.

[12] S. Liaw and et al. A renormalization group framework for scale-invariant feature learning in deep neural networks. In *AAAI Conf. on Artificial Intelligence*, 2025.

[13] Paras P. Mehta and D. J. Schwab. Renormalization group and deep learning. arXiv:1410.3831, 2014.

[14] Lars Onsager. Crystal statistics i: A two-dimensional model with an order-disorder transition. *Phys. Rev.*, 65:117–149, 1944.

[15] Tao Pang and et al. Neural renormalization group. *Phys. Rev. E*, 101:063301, 2020.

[16] Stephen B. Pope. *Turbulent Flows*. Cambridge University Press, 2000.

[17] Joseph Smagorinsky. General circulation experiments with the primitive equations. *Mon. Weather Rev.*, 91:99–164, 1963.

[18] H. Eugene Stanley. *Introduction to Phase Transitions and Critical Phenomena*. Oxford University Press, 1971.

[19] Kenneth G. Wilson. The renormalization group. *Rev. Mod. Phys.*, 47:773–840, 1974.

[20] Kenneth G. Wilson and J. Kogut. The renormalization group and the ϵ expansion. *Phys. Rep.*, 12:75–200, 1975.

[21] Chiyuan Zhang and et al. Understanding deep learning requires rethinking generalization, 2016. arXiv:1611.03530.

[22] Jean Zinn-Justin. *Quantum Field Theory and Critical Phenomena*. Oxford University Press, 5th edition, 2021.